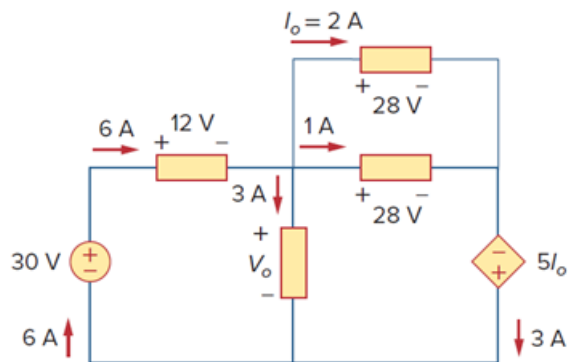


Problem

Find V_o and the power absorbed by each element in the circuit of Fig. 1.31.

Figure 1.31:



Step-by-step solution

Step 1 of 5

Refer to the circuit diagram depicted in Figure 1.31 in the textbook.

Re-draw the circuit diagram by labelling with powers to all the circuit elements.

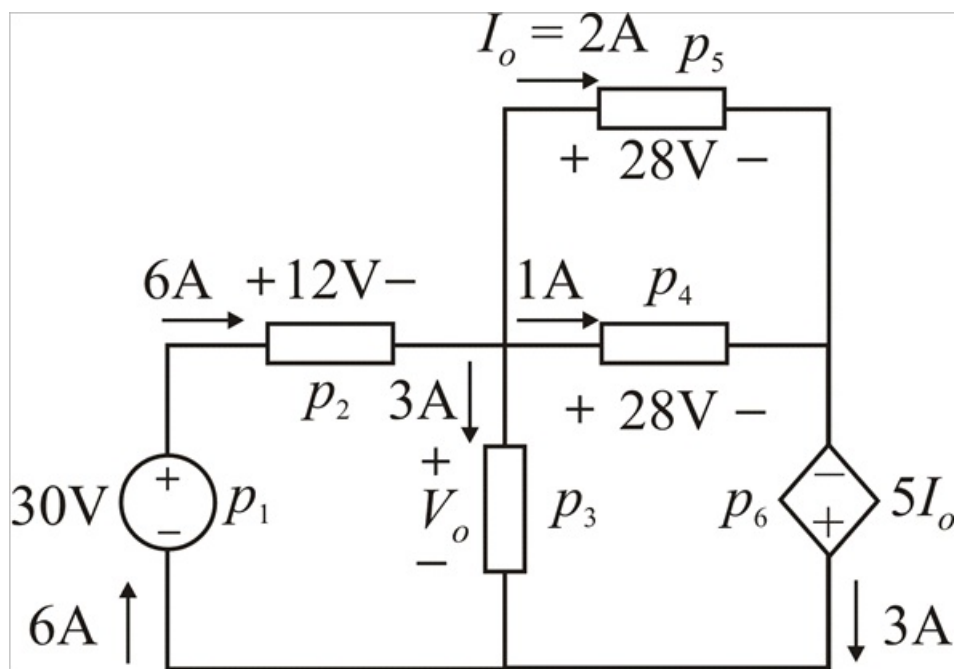


Figure 1

Step 2 of 5

If the current is entering into the positive terminal, the power absorbed is positive (+ve). If the current is entering into the negative terminal, the power absorbed is negative (-ve).

For power p_1 , 6 A current is coming out of the positive terminal of the voltage source (or into the negative terminal). Therefore,

$$\begin{aligned} p_1 &= 30(-6) \\ &= -180 \text{ W} \end{aligned}$$

For power p_2 , 6 A current is coming out of the negative terminal (or into the positive terminal). Therefore,

$$\begin{aligned} p_2 &= 12(6) \\ &= 72 \text{ W} \end{aligned}$$

Step 3 of 5

For power p_3 , 3 A current is coming out of the negative terminal (or into the positive terminal). Therefore,

$$\begin{aligned} p_3 &= V_o(3) \\ &= 3V_o \text{ W} \end{aligned}$$

For power p_4 , 1 A current is coming out of the negative terminal (or into the positive terminal). Therefore,

$$\begin{aligned} p_4 &= 28(1) \\ &= 28 \text{ W} \end{aligned}$$

Step 4 of 5

For power p_5 , 2 A current is coming out of the negative terminal (or into the positive terminal). Therefore,

$$\begin{aligned} p_5 &= 28(2) \\ &= 56 \text{ W} \end{aligned}$$

For power p_6 , 3 A current is coming out of the positive terminal (or into the negative terminal). Therefore,

$$\begin{aligned} p_6 &= (5I_o)(-3) \\ &= [5(2)](-3) \\ &= -30 \text{ W} \end{aligned}$$

Step 5 of 5

The algebraic sum of powers in a circuit, at any instant of time should be zero. Therefore,

$$\begin{aligned} p_1 + p_2 + p_3 + p_4 + p_5 + p_6 &= 0 \\ -180 + 72 + 3V_o + 28 + 56 - 30 &= 0 \\ 3V_o - 54 &= 0 \end{aligned}$$

Simply the expression further to find the output voltage, V_o .

$$\begin{aligned} 3V_o &= 54 \\ V_o &= \frac{54}{3} \\ &= 18 \text{ V} \end{aligned}$$

Therefore, the value of output voltage, V_o in the circuit is 18 V.